

Sameer Aryal

CONTACT INFORMATION	Stanley Center for Psychiatric Research Broad Institute of MIT and Harvard 75 Ames St Cambridge, MA 02142	Phone: (413) 347-9726 E-mail: sameer.aryal@gmail.com LinkedIn: https://www.linkedin.com/in/sameer-aryal-nyu/ Website: www.sameeraryal.com
RESEARCH INTERESTS	Genome-scale neurogenetics; high-throughput examination of the mechanisms of gene expression in neural systems.	
EDUCATION	New York University , New York, NY Doctor of Philosophy, Basic Medical Science, August 2013 – May 2020 • Advisor: Eric Klann, Ph.D. • Dissertation: “Molecular and computational examination of <i>de novo</i> protein synthesis in fragile X syndrome” Williams College , Williamstown, MA Bachelor of Arts, Biology (honors) and Economics, September 2008 – June 2012 • Advisor: Tim J. Lebestky, Ph.D. • Honors Thesis: “The role of DopR neuronal circuits in regulating endogenous arousal in <i>D. melanogaster</i>”	
RESEARCH EXPERIENCE	<i>Stanley Center for Psychiatric Research, Broad Institute</i> , Cambridge, MA <i>Research Scientist</i> January 2024 – present Lead a team conducting computational analyses and functional experiments on schizophrenia mouse models to drive neurobiological insights and advance therapeutic target identification. <i>Postdoctoral Associate</i> , Advisor: Morgan Sheng, M.D., Ph.D. October 2020 – December 2023 Examined molecular and functional alterations in brains of humans with schizophrenia and bipolar and in mouse models of these disorders, employing high-throughput proteomic, transcriptomic, and computational approaches to identify disease-associated molecular and functional alterations. <i>Center for Neural Science, New York University</i> , New York, NY <i>Graduate Research Assistant</i> , Advisor: Eric Klann, Ph.D. June 2014 – June 2020 Discovered a mechanism underlying elevated mRNA translation in a mouse model of fragile X syndrome by designing and executing molecular and genome-wide assays (Ribo-Seq, Trap-Seq), performing all computational analyses of high-throughput data, and developing a novel approach to measure ribosome translocation rates in primary neurons. <i>Neuroscience Research Partnership, A*STAR - Duke University - NUS</i> , Singapore <i>Research Assistant</i> , Advisor: Adam Claridge-Chang, Ph.D. August 2012 – May 2013 Designed and implemented MATLAB pipelines to analyze <i>D. melanogaster</i> motion-tracking data and developed a modular library for generating estimation plots that emphasize effect sizes over p-values. Awarded the Singapore International Pre-Graduate Award (SIPGA) for this work. <i>Department of Biology, Williams College</i> , Williamstown, MA <i>Honors Student</i> , Advisor: Tim J. Lebestky, Ph.D. September 2011 – May 2012 Investigated neuronal circuits regulating arousal and their connectivity in the <i>D. melanogaster</i> brain by conditionally activating neurons expressing temperature-gated trpA1 channels, followed by behavioral analysis using sleep and locomotor assays.	

1. Farsi, Z., Nicoletta, A., Simmons, S. K., **Aryal, S.**, Shepard, N., Brenner, K., Lin, S., Herzog, L. E., Moran, S. P., Stalnaker, K. J., et al. Brain region-specific changes in neurons and glia and dysregulation of dopamine signaling in Grin2a mutant mice. *Neuron*, 111(21):3378–3396, 2023.
2. **Aryal, S.**, Bonanno, K., Song, B., Mani, D. R., Keshishian, H., Carr, S. A., Sheng, M., Dejanovic, B. Deep proteomics identifies shared molecular pathway alterations in synapses of schizophrenia and bipolar disorder patients and mouse model. *Cell Reports*, 42(5), 2023.
3. Longo, F., **Aryal, S.**, Anastasiades, P. G., Maltese, M., Baimel, C., Albanese, F., Tabor, J., Zhu, J., Oliveira, M. M., Gastaldo, D., et al. Cell-type-specific disruption of cortico-striatal circuitry drives repetitive patterns of behavior in fragile X syndrome model mice. *Cell Reports*, 42(8), 2023.
4. **Aryal, S.**, Longo, F., Klann, E. Genetic removal of p70 S6K1 corrects coding sequence length-dependent alterations in mRNA translation in fragile X syndrome mice. *Proceedings of the National Academy of Sciences USA*, 118(18):e2001681118, 2021.
5. Longo, F., Mancini, M., Ibraheem, P. L., **Aryal, S.**, Mesini, C., Patel, J. C., Penhos, E., Rahman, N., Donohue, M., Santini, E., et al. Cell-type-specific disruption of PERK-eIF2 α signaling in dopaminergic neurons alters motor and cognitive function. *Molecular Psychiatry*, 26(11):6427–6450, 2021.
6. Ho, J., Tumkaya, T., **Aryal, S.**, Choi, H., Claridge-Chang, A. Moving beyond P values: data analysis with estimation graphics. *Nature Methods*, 16(7):565–566, 2019.
7. Bowling, H., Bhattacharya, A., Zhang, G., [et al., incl. **Aryal, S.**], Klann, E. Altered steady state and activity-dependent de novo protein expression in fragile X syndrome. *Nature Communications*, 10(1):1710, 2019.
8. **Aryal, S.**, Klann, E. Turning up translation in fragile X syndrome. *Science*, 361(6403):648–649, 2018.
9. Mohammad, F., **Aryal, S.**, Ho, J., Stewart, J. C., Norman, N. A., Tan, T. L., Eisaka, A., Claridge-Chang, A. Ancient anxiety pathways influence Drosophila defence behaviors. *Current Biology*, 26(7):981–986, 2016.
10. Song, B. J., Ge, Y., Nicoletta, A., Kwon, M. J., Lodder, B., Bonanno, K., Valle-Tojeiro, A., Adeleye, J., Geng, C., Natarajan, S., et al. Elevated synaptic PKA activity and abnormal striatal dopamine signaling in Akap11 mutant mice, a genetic model of schizophrenia and bipolar disorder. *bioRxiv*, 2024.
11. Kwon, M. J., Perzel Mandell, K. A., **Aryal, S.**, Geng, C., Nicoletta, A. A., Valle-Tojeiro, A., Natarajan, S., Misri, D., Hall, A., Song, B. J., et al. Multi-omics characterization reveals brain-wide disruption of synapses and region- and age-specific changes in neurons and glia in Sp4 mutant mice, a genetic model of schizophrenia and bipolar disorder. *bioRxiv*, 2024.
12. **Aryal, S.**, Geng, C., Kwon, M. J., Farsi, Z., Goble, N., Asan, A. S., Brenner, K., Shepard, N., Seidel, O., Wang, Y., et al. Reduction of SynGAP- γ , disrupted splicing of Agap3, and oligodendrocyte deficits in Srrm2 mice, a genetic model of schizophrenia and neurodevelopmental disorder. *bioRxiv*, 2024.
13. Falkovich, R., **Aryal, S.**, Wang, J., Sheng, M., Bathe, M. Synaptic composition, activity, mRNA translation and dynamics in combined single-synapse profiling using multimodal imaging. *bioRxiv*, 2024.
14. Huang, W.-C., Kast, R., Mandell, K. P., Dejanovic, B., Bonanno, K., **Aryal, S.**, Farsi, Z., Wilde, J., Wang, D., Gao, X., et al. Brain-region-specific changes and dysregulation of activity regulated genes in Gria3 mutant mice, a genetic animal model of schizophrenia. *bioRxiv*, 2024.
15. Huang, W.-C., Mandell, K. P., **Aryal, S.**, Song, B. J., Valle-Tojeiro, A., Goble, N., Geng, C., Asan, A. S., Liu, X.-M., Dennis, C., et al. Epigenetic changes, neuronal dysregulation

and metabolomic abnormalities in Zmym2 mutant mice, a genetic model of schizophrenia and neurodevelopmental disorders. *bioRxiv*, 2025.

16. Mandell, K. A. P., Simmons, S. K., Nadig, A., Farsi, Z., Huang, W.-C., **Aryal, S.**, Kwon, M. J., Song, B., Brenner, K., Shepard, N., et al. Meta-analysis of the brain transcriptomes of multiple genetic mouse models of schizophrenia highlights dysregulation in striatum and thalamus. *bioRxiv*, 2025.

HONORS AND AWARDS

- Awarded *registration and travel scholarship* to attend the 2017 Summer Institute in Statistics for Big Data, University of Washington, Seattle. *April 2017*
- Inducted into the *Sigma Xi Scientific Research Society*. *June 2012*
- Recipient of the *Robert L. Gaudino Fellowship* at Williams College to conduct independent Economics research on healthcare access in rural Nepal. *January 2010*
- Awarded *Brilliance in Nepal* by the British Council, Kathmandu, for achieving the highest marks in University of Cambridge Advanced-Level Biology examinations across Nepal, May/June 2007 session. *April 2008*
- Graduated *Valedictorian* from Budhanilkantha School, the national school of Nepal. *May 2007*

CONFERENCE PRESENTATIONS

- *Invited speaker*, Nanosymposium on “Neurodevelopmental Disorders - Mechanisms.” Society for Neuroscience Annual Meeting. 2017.
- *Invited speaker*, Annual Molecular Pharmacology Retreat. Sackler Institute at NYU School of Medicine. 2017.
- Presented posters describing my graduate research at:
 - Society for Neuroscience Annual Meeting. 2015, 2016, and 2018.
 - Molecular and Cellular Cognition Society Annual Meeting. 2015, 2016, and 2018.
 - Brains and Behavior: Order and Disorder in the Nervous System. Cold Spring Harbor Laboratories Symposium in Quantitative Biology. 2018.
 - RNA Therapeutics: From Base Pairs to Bedsides Symposium. 2018.

ADDITIONAL QUALIFICATIONS

- *Intern* at the Office of Industrial Liaison, NYU Langone Health. Created a business plan for an NYU-developed technology by researching the technology, identifying its competitive advantages, analyzing the market for the technology, and describing the business model for the company. *July - Nov 2017*
- *Graduate Teaching Assistant* to Dr. Angus Wilson in the seminar “Introduction to Research” at the Sackler Institute at NYU School of Medicine. *Fall 2017*
- *Peer tutor* for beginner and intermediate Macroeconomics and Econometrics at Williams College, MA. *2010-2012*

SKILLS

- Programming languages: Python, R, MATLAB, Shell scripting (Bash).
- Applications: Git, L^AT_EX, Jupyter notebook.
- Environments: High Performance Computing (SGE/Slurm), Unix/Linux, Windows.
- Assays: Transcriptomics (single-nucleus and bulk RNA-Seq), translomics (ribosome profiling and Trap-seq), Proteomics (MS/MS of purified synapses).
- Laboratory techniques : Primary neuron culture, transgene delivery with AAV and lentiviral vectors, in vivo pharmacology in mice, nucleic acid purification, high-throughput library preparations, qPCR, immuno-precipitation, western blotting, immuno-fluorescence, confocal microscopy, immunological assay development.